# Video Transcript

## Video 5: Introduction to Bayes Theorem (03:22)

So, are rich people happy? What do you think? A Harvard MBA student told us that only 10% of happy people are rich. Well, let's see if we can figure out what that means. We're going to talk about Bayes theorem. So, here's the background for it and the way that I remember it. So, in set theory, we've got two sets A and B, and their intersection is written as A intersection B. So, here are those two sets, and this is A intersection B. It's the area that's common to A, both A and B are in this.

Okay, so that's the mental model, and we can approach it two ways. We can say let's suppose A and B are events. So, these are both A and B have occurred if we're in this middle region. So, this is where event A has occurred and this is where event B has occurred. And then this would be if A has occurred and then B occurs, or if B has occurred and then A occurs. Let's take a look to see how we can write this. So, in probability notation, we say the probability of event A is P(A). So, that would be the area of the circle A. The probability of event B is P(B). Now the probability of B, given that A has already occurred, is written as P, B and then the vertical line A. And we read this as we're saying, this is the conditional probability that B occurs given that A has occurred, or the probability of B given A. So, the probability of A and B is probability of B given A. But that must be the same as the probability of A occurring given that B has occurred times the probability of B because they're both A intersection B.

So, this is Bayes theorem. Okay? That it's A intersection B and we can write it two ways, the probability of A given B and then the probability of B. And the probability of B given A times the probability of A. And just rearranging this, we get that the probability of B given A is the probability of A given B times the probability of B divided by the probability of A. Now this doesn't seem too useful. It just looks like trickery, but in fact it's very useful. So, let's go and see if we can solve the problem of are rich people happy. Bye for now.

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